Synthetic Cold Splits Are Not Enough: Distribution-Shift Benchmarking for Drug-Target Interaction Prediction

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Abstract

Drug-target interaction (DTI) papers often use synthetic cold-start splits as proxies for generalization, but it remains unclear whether conclusions drawn from these splits transfer to real distribution shifts. We present a benchmark-first case study showing that, within a studied three-model panel, rankings are sensitive to how the shift is defined. Using a compact external panel consisting of a fully specified fixed-feature `base` encoder and two recent literature baselines, `DTI-LM` and `HyperPCM`, we first build a 3-seed synthetic reference panel on `BindingDB_Kd`. Even within this single dataset, rankings are heterogeneous: `HyperPCM` is strongest on synthetic `unseen_drug` by mean AUPRC, `blind_start` has no reliable winner under paired bootstrap, and `base` remains strongest on synthetic `unseen_target`. We then evaluate the same panel on a real temporal/provenance benchmark, `BindingDB_patent / patent_temporal`, where the winner changes again. Over 3 seeds, `DTI-LM` becomes the strongest model at 0.7825 ± 0.0078 AUPRC, ahead of `base` (0.7772 ± 0.0086) and `HyperPCM` (0.7753 ± 0.0050). This margin is statistically positive but small in absolute magnitude, so we interpret it as a panel-level ranking change rather than a large deployment gap. An earlier alternative patent cutoff and a non-patent temporal probe preserve the same `DTI-LM > base` ordering. A pooled year-band and overlap-bucket diagnosis further shows that temporal/provenance OOD is structured rather than monolithic: different models dominate different temporal slices and overlap strata. We package the split manifests, benchmark resources, and standalone evaluator for reuse. Within the studied model panel and benchmarks, these results show that synthetic cold splits and real OOD shifts should not be treated as interchangeable evaluation settings for DTI.

1 Introduction

Predicting drug-target interactions for unseen molecules or proteins is a central goal in computational drug discovery. Because truly prospective evaluation is difficult and expensive, many recent studies report held-out drug, target, or blind splits on datasets derived from BindingDB or DAVIS as proxies for inductive generalization [1, 2, 3, 4, 5, 6, 7, 8]. These protocols are useful, but they also encourage a strong implicit assumption: if a method performs well on one cold-start split, then it is broadly robust to the kinds of shift encountered in deployment.

That assumption is weaker than it appears. Different split definitions remove different forms of overlap, and they may favor different model families. A pooled-LM baseline that looks strong under one synthetic exclusion rule may no longer be the most reliable option once the benchmark shifts in time, provenance, or assay composition. If these ranking changes occur even within a compact studied panel, benchmark design is not a technical footnote. It becomes part of the scientific claim.

This paper studies that question directly. Rather than proposing another universal architecture, we ask whether the ranking of a compact DTI model panel is preserved across synthetic cold-start regimes and real distribution shifts. The goal is to test ranking stability under benchmark shift in a controlled case study, not to declare a universal field-wide leaderboard. Our starting point is a synthetic reference panel on `BindingDB_Kd`, where we compare three representative systems: a fixed-feature MLP reference

model (base) and two recent external literature baselines, DTI-LM and HyperPCM. Even here, rankings are already heterogeneous: HyperPCM is strongest on synthetic unseen_drug by mean AUPRC, blind_start has no reliable winner under paired bootstrap, and base remains strongest on synthetic unseen_target.

We then contrast this synthetic picture with a real temporal/provenance benchmark built from the BindingDB patent split (patent_temporal), using matched 3-seed evaluation. The result is another winner change. The synthetic panel does not point to a single dominant system, but the promoted patent benchmark clearly favors DTI-LM by mean AUPRC. Paired bootstrap confidence intervals support the headline temporal comparison: DTI-LM - base remains positive on patent temporal. An earlier-cutoff variant of the patent benchmark (patent_temporal_v2017) preserves the same DTI-LM > base ordering over matched three-seed evaluation.

To move beyond leaderboard-style reporting, we also diagnose the promoted temporal benchmark by year and overlap bucket. This analysis shows that temporal OOD is not simply "harder cold-start." Instead, the gap is concentrated in specific year bands and overlap regimes, which helps explain why a method that looks strong under one split can lose under another. We further include matched 3-seed support on BindingDB_Ki and DAVIS to test whether the same external panel remains stable across shifted target-cold settings. Recent molecular benchmark studies have made analogous arguments about OOD sensitivity in adjacent tasks [9, 10, 11]; our contribution is to show the same issue concretely in DTI and to release the benchmark suite as a reusable split-plus-evaluator resource for reliable DTI evaluation.

Our contributions are fivefold. First, we construct a 3-seed synthetic reference panel showing that even standard cold-start evaluation is regime-dependent. Second, we establish a 3-seed real-OOD benchmark on BindingDB_patent and show a ranking reversal relative to the synthetic reference panel. Third, we expand the model panel with recent pooled-LM baselines and show that the strongest promoted recent baseline is benchmark-dependent rather than uniformly dominant within the studied axes. Fourth, we provide year-band and overlap-bucket analyses that connect the ranking change to measurable shift axes, matched 3-seed support on BindingDB_Ki and DAVIS, and two review-driven robustness probes. Fifth, we release canonical split manifests, reusable benchmark resources, and a standalone evaluator to make the suite easy to reuse.

2 Related Work

Synthetic cold-start evaluation in DTI. Cold-start evaluation has become standard practice in DTI and drug-target affinity prediction. Recent models routinely report random, held-out drug, held-out target, or blind-start splits on BindingDB-derived and DAVIS-derived benchmarks [3, 4, 5, 6, 7, 8]. This literature established synthetic cold-start testing as a useful proxy for inductive generalization. However, it often treats different split definitions as variations of the same evaluation problem. Our results suggest that this assumption is too coarse, because different synthetic regimes already produce different model rankings.

Retrieval, inductive, and memory-based DTI models. Another line of work designs architectures intended to transfer information from seen entities to unseen drugs or targets. Retrieval-based or transfer-oriented models attempt to reuse evidence from related compounds and proteins, while inductive or prototype-like systems aim to improve robustness under entity holdout [4, 5, 6, 7, 8]. Our work does not claim that these mechanisms are ineffective. On the contrary, we recover setting-specific gains for retrieval on synthetic blind_start and for target-memory on synthetic unseen_target. The central point is different: benchmark choice determines whether these gains appear.

Table 1: Benchmark taxonomy and seed0 split statistics. Test prevalence differs substantially across benchmark families, so AUPRC should be compared within benchmark rather than across rows.

Benchmark	Shift family	Synth.?	Test rows	Pos. rate
`Kd / unseen_drug`	drug-cold	yes	12,744	0.1966
`Kd / blind_start`	drug+target-cold	yes	2,821	0.2031
`Kd / unseen_target`	target-cold	yes	12,052	0.2239
`patent / patent_temporal`	temporal+source	no	49,028	0.5745
`Ki / unseen_target`	assay+target-cold	mixed	70,380	0.4652
`DAVIS / unseen_target`	dataset+target-cold	mixed	5,168	0.0615

Distribution shift and benchmark design. Recent benchmark studies in molecular machine learning and related biomedical prediction problems have shown that model rankings can change under scaffold, temporal, or domain-shifted evaluation [9, 10, 11]. Our paper brings that diagnostic perspective into DTI. Unlike a leaderboard paper, we intentionally keep the model panel compact. The main result is not that one architecture dominates many baselines, but that even a small, representative panel is enough to expose ranking instability across benchmark definitions.

Positioning. Prior DTI papers typically optimize for one split regime or propose a stronger model within a fixed benchmark setup. This paper instead studies whether model rankings remain stable across synthetic cold-start regimes and a real-OOD patent benchmark, then diagnoses the shift structure behind disagreements.

3 Benchmark Setup

Our goal is to test whether a compact DTI model panel preserves its ranking across synthetic cold-start regimes and real distribution shifts. The replacement panel used in the revised manuscript contains three models: `base`, a fixed-feature MLP reference system built from RDKit Morgan fingerprints and hashed protein k-mer features; `DTI-LM`, a pooled ChemBERTa+ESM2 baseline aligned to recent literature; and `HyperPCM`, a task-conditioned pooled-LM baseline. We intentionally keep this external panel small so that benchmark effects are not obscured by a large architecture zoo while still avoiding dependence on internal probe models. Because the anchor system is central to every ranking claim, Appendix Table A3 gives its exact feature construction, architecture, parameter count, and training recipe.

We evaluate this panel on two benchmark groups. The first is a synthetic reference panel based on `BindingDB_Kd`, with three split definitions: `unseen_drug`, `blind_start`, and `unseen_target`. The second is a real-OOD screening panel. Our primary real-OOD benchmark is the BindingDB patent split (`patent_temporal`), which uses a year-based temporal partition and is evaluated over three seeds. We also include `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting shifted target-cold benchmarks that test whether the same external model family remains stable across assay and dataset shifts. Table 1 summarizes the shift taxonomy and split statistics.

We report AUPRC as the primary metric and AUROC as a secondary metric. AUPRC is emphasized because class balance and ranking behavior are central to the benchmark claims in this paper. For the replacement panel, we report matched three-seed results on the synthetic reference panel and the promoted patent benchmark. We first run a transparent seed0 screening pass on `patent_temporal`, `blind_start`, and `unseen_target`, then promote the strongest or most informative recent baselines to matched three-seed evaluation. `DTI-LM` is promoted because it is the strongest seed0 and 3-seed recent baseline on the primary

temporal/provenance benchmark patent_temporal, while HyperPCM is kept because it is competitive on the synthetic reference panel and therefore provides a meaningful replacement for the internal probe lines. Appendix Table A7 makes this promotion logic explicit. We further add paired bootstrap uncertainty support for the headline external-panel comparisons against base, including the promoted patent comparison and the synthetic reference-panel contrasts where matched three-seed results are available. Appendix Table A6 collects these intervals.

To make the benchmark reusable, each split is exported as a canonical resource bundle with manifests, split tables, cached identifiers, and a standalone CSV-based evaluator. This resource layer allows external predictions to be scored against the exact same test partitions used in the paper. Appendix Table A8 enumerates the released artifacts and how they are consumed by the evaluator.

The patent benchmark requires several setup choices that are documented explicitly. We use a local year-based split of BindingDB_patent, with training years before the validation year, a held-out validation year, and a later temporal test set. Because this benchmark also changes provenance and assay composition, we describe it as temporal/provenance OOD rather than pure time shift. We also use a patent-specific affinity conversion rather than reusing the standard pKd transform, because the patent benchmark stores affinities in a different log-scale convention. Appendix Table A1 summarizes the exact year boundaries, split counts, label construction, overlap policy, and seed semantics for both patent_temporal and the alternative-cutoff patent_temporal_v2017. Appendix Table A2 provides the same information for the review-driven BindingDB_nonpatent_Kd / nonpatent_temporal probe. BindingDB-derived panels are anchored in the BindingDB resource [1], while DAVIS provides an external kinase-focused support benchmark [2].

Finally, for diagnostic analysis, we use the saved overlap score from each test prediction to compute overlap buckets that summarize how strongly each test example resembles the training support. Concretely, overlap_score is defined as $\min(s_d, s_t)$, where s_d is the cosine similarity between the test drug's Morgan fingerprint and its nearest training-drug neighbor, and s_t is the cosine similarity between the test target's hashed k-mer feature vector and its nearest training-target neighbor. We use the minimum to conservatively define overlap: a pair is high-overlap only when both its drug and target are well represented in the training set. We retain cosine for the released overlap score so that drug-side and target-side overlap share the same similarity scale inside this joint diagnostic; the separate synthetic similarity analysis still reports the standard Morgan-fingerprint Tanimoto and scaffold-overlap statistics. We report benchmark taxonomy and split statistics, including test-set size and prevalence, so that AUPRC is interpreted within benchmark rather than across benchmark families.

4 Results

4.1 Synthetic cold splits are already regime-dependent

We begin with the BindingDB_Kd synthetic reference panel. Table 2 shows the main pattern: even within a single dataset, model rankings are not stable across split definitions. On unseen_drug, HyperPCM is strongest by mean AUPRC, reaching 0.5694 ± 0.0473 versus 0.5509 ± 0.0900 for base and 0.5004 ± 0.0341 for DTI-LM. On blind_start, HyperPCM has the highest mean AUPRC at 0.3828 ± 0.1024, but the margin over base is much smaller, the variance is the largest in the main table, and paired bootstrap crosses zero; we therefore treat this row as having no reliable winner. On unseen_target, the pattern changes again: base is strongest by mean AUPRC at 0.5224 ± 0.0602, while DTI-LM is slightly lower on AUPRC but higher on mean AUROC.

These results matter because they show that "synthetic cold-start DTI" is already not a single evaluation condition. Even after replacing internal probes with external literature baselines, different synthetic regimes still reward different model families, so it is unsafe to generalize from one split to another without explicit

Table 2: Synthetic reference panel and promoted real-OOD benchmark under the external replacement panel. Values are mean ± std over seeds. Best mean AUPRC in each row is bold. The patent delta is statistically supported but small in absolute magnitude, so row-wise winner labels should be read as within-panel ordering rather than deployment-sized gaps.

Benchmark	Seeds	base (AUPRC/AUROC)	DTI-LM (AUPRC/AUROC)	HyperPCM (AUPRC/AUROC)	Note	Winner
`BindingDB_Kd / unseen_drug`	3	0.5509 ± 0.0900 / 0.7794 ± 0.0534	0.5004 ± 0.0341 / 0.7364 ± 0.0122	**0.5694 ± 0.0473 / 0.7778 ± 0.0448**	synthetic drug-cold favors `HyperPCM` by mean AUPRC	HyperPCM
`BindingDB_Kd / blind_start`	3	0.3737 ± 0.0229 / 0.6778 ± 0.0151	0.3636 ± 0.0693 / 0.6490 ± 0.0642	**0.3828 ± 0.1024 / 0.6807 ± 0.0824**	`HyperPCM` has the highest mean AUPRC, but paired bootstrap crosses zero so this row has no reliable winner	none
`BindingDB_Kd / unseen_target`	3	**0.5224 ± 0.0602 / 0.7741 ± 0.0161**	0.5136 ± 0.0730 / 0.7970 ± 0.0159	0.4955 ± 0.0793 / 0.7810 ± 0.0191	synthetic target-cold reverts to `base` on mean AUPRC	base
`BindingDB_patent / patent_temporal`	3	0.7772 ± 0.0086 / 0.7223 ± 0.0109	**0.7825 ± 0.0078 / 0.7362 ± 0.0083**	0.7753 ± 0.0050 / 0.7368 ± 0.0090	promoted temporal/provenance benchmark gives a small but consistent `DTI-LM` edge	DTI-LM

evidence. The review-driven similarity diagnosis sharpens this point further: on synthetic `unseen_drug`, the held-out compounds still have median nearest-train Tanimoto 0.7732 and Murcko scaffold match rate 0.6204. When we replace entity holdout with a stricter scaffold split, the earlier `HyperPCM` edge is no longer supported. Over three seeds on `BindingDB_Kd / scaffold_drug`, `base` reaches $0.6069 \pm 0.0161/0.8351 \pm 0.0037$, while `HyperPCM` reaches $0.5983 \pm 0.0368/0.8155 \pm 0.0204$, and the paired bootstrap interval for `HyperPCM - base` crosses zero (Appendix Tables A5 and A6). The synthetic panel therefore remains useful, but its drug-cold ordering is sensitive to how chemically strict the split actually is.

4.2 Real temporal shift favors a different strong baseline

We next evaluate the same external panel on `BindingDB_patent / patent_temporal`, our primary real-OOD benchmark. Over three seeds, `DTI-LM` is the strongest model by mean AUPRC, reaching 0.7825 ± 0.0078, compared with 0.7772 ± 0.0086 for `base` and 0.7753 ± 0.0050 for `HyperPCM`. Figure 1 compresses the main benchmark story into within-benchmark AUPRC deltas relative to `base`: synthetic cold splits favor different systems, while the promoted temporal/provenance benchmark favors `DTI-LM`. This advantage is also supported by paired bootstrap: `DTI-LM - base` $= +0.0054$ with 95% CI $[+0.0028, +0.0079]$. At the same time, the absolute margin is small relative to the models' own seed variation, so we interpret it as a ranking reversal within the studied panel rather than as a large deployment-relevant gap by itself. A second temporal robustness probe strengthens this reading. On the non-patent benchmark built directly from the official BindingDB release, `DTI-LM` still beats `base` over matched 3-seed evaluation, reaching $0.6531 \pm 0.0066/0.7644 \pm 0.0041$ versus $0.6369 \pm 0.0173/0.7333 \pm 0.0081$ for `base` (Appendix Table A5). This does not make the patent benchmark a pure time-only shift, but it does reduce the concern that the main winner change is driven only by patent provenance.

This is the central winner change in the revised paper. The synthetic panel does not produce a single best model, but the promoted temporal/provenance benchmark favors `DTI-LM` by mean AUPRC. The alternative-cutoff benchmark tells the same story over matched three-seed evaluation: on `patent_temporal_v2017`, `DTI-LM` reaches $0.7134 \pm 0.0027/0.6754 \pm 0.0039$, versus $0.7044 \pm 0.0058/0.6624 \pm 0.0079$ for `base`. The broader lesson is therefore not that one external baseline wins everywhere, but that both benchmark axis and model family change the apparent ranking within the studied panel.

4.3 Temporal shift is structured across year bands and overlap levels

To help explain why the temporal/provenance benchmark changes the ranking, we analyze the patent results by year band and by overlap bucket. Table 3 makes the primary AUPRC slices explicit, while Figure 2 visualizes the same structure. In the large 2019 slice, `DTI-LM` is strongest by mean AUPRC (0.7734 ± 0.0071), with `base` (0.7626 ± 0.0098) and `HyperPCM` (0.7637 ± 0.0049) close behind. To avoid over-reading the tiny 2021 cell, we pool 2020 and 2021 into a single recent stratum. In that pooled recent slice, the ordering flips: `base` reaches $0.8716 \pm 0.0121/0.8134 \pm 0.0133$, while `DTI-LM` and `HyperPCM` drop to $0.8533 \pm 0.0100/0.7938 \pm 0.0166$ and $0.8516 \pm 0.0139/0.7925 \pm 0.0209$, respectively.

The overlap-bucket analysis reveals another useful pattern. In the lowest-overlap bucket, `DTI-LM` is strongest by mean AUPRC (0.6353), while `HyperPCM` is close behind. In the middle-overlap bucket,

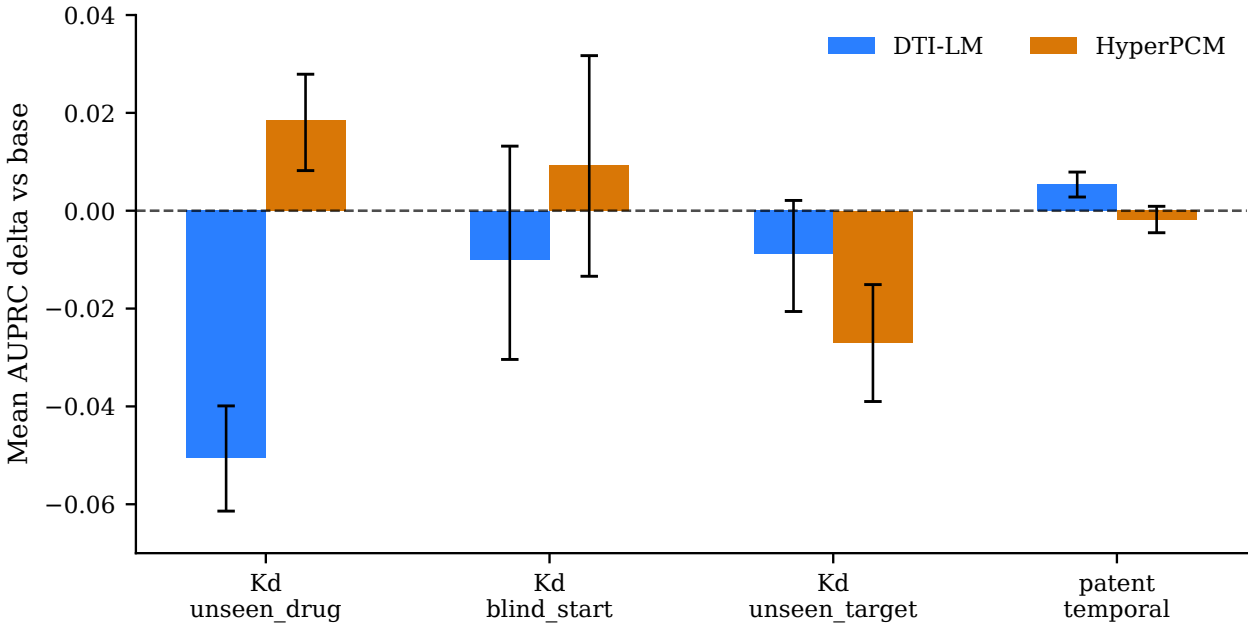

Figure 1: Mean within-benchmark AUPRC delta relative to `base` under the external replacement panel. Error bars show paired bootstrap 95% intervals where available. Because prevalence differs across benchmarks, the vertical scale should be read as a row-wise within-benchmark contrast rather than as a cross-benchmark effect-size scale.

Table 3: Compact AUPRC summary behind the patent diagnosis under the external replacement panel. Values are mean ± seed std over the matched 3-seed runs. The tiny 2021 slice is pooled into the `2020-2021` stratum to avoid over-interpreting a 72-example cell.

Slice type	Slice	Count	base	DTI-LM	HyperPCM
Year	2019	42,009	0.7626 ± 0.0098	$\mathbf{0.7734} \pm \mathbf{0.0071}$	0.7637 ± 0.0049
Year	2020–2021	7,019	$\mathbf{0.8716} \pm \mathbf{0.0121}$	0.8533 ± 0.0100	0.8516 ± 0.0139
Overlap bucket	0 (0.000–0.531)	16,341	0.5551 ± 0.0190	$\mathbf{0.6353} \pm \mathbf{0.0175}$	0.6147 ± 0.0208
Overlap bucket	1 (0.531–0.710)	16,342	0.7218 ± 0.0167	0.7489 ± 0.0094	$\mathbf{0.7513} \pm \mathbf{0.0060}$
Overlap bucket	2 (0.710–0.989)	16,345	$\mathbf{0.9237} \pm \mathbf{0.0030}$	0.8866 ± 0.0097	0.8767 ± 0.0042

`HyperPCM` takes a small lead on AUPRC (0.7513) over `DTI-LM` (0.7489) and `base` (0.7218). In the highest-overlap bucket, the ordering flips again and `base` dominates on both AUPRC and AUROC. Together, these results are consistent with the view that temporal OOD is not just a colder version of synthetic cold-start. Instead, it changes which subpopulations dominate the aggregate result, and therefore changes the model ranking.

4.4 Supporting target-cold OOD benchmarks still favor the plain base model

We treat `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting real-OOD benchmarks rather than primary headline axes. Under matched three-seed evaluation, neither promoted external baseline overturns the support-panel story. On `BindingDB_Ki`, `base` reaches $0.6934 \pm 0.0257/0.7192 \pm 0.0068$, compared with $0.6648 \pm 0.0177/0.6936 \pm 0.0044$ for `DTI-LM` and $0.6515 \pm 0.0301/0.6757 \pm 0.0211$ for `HyperPCM`. On `DAVIS`, `base` reaches $0.4097 \pm 0.0275/0.8613 \pm 0.0153$, compared with $0.4090 \pm 0.0570/0.8718 \pm 0.0152$ for `DTI-LM` and $0.3840 \pm 0.0269/0.8610 \pm 0.0245$ for `HyperPCM`.

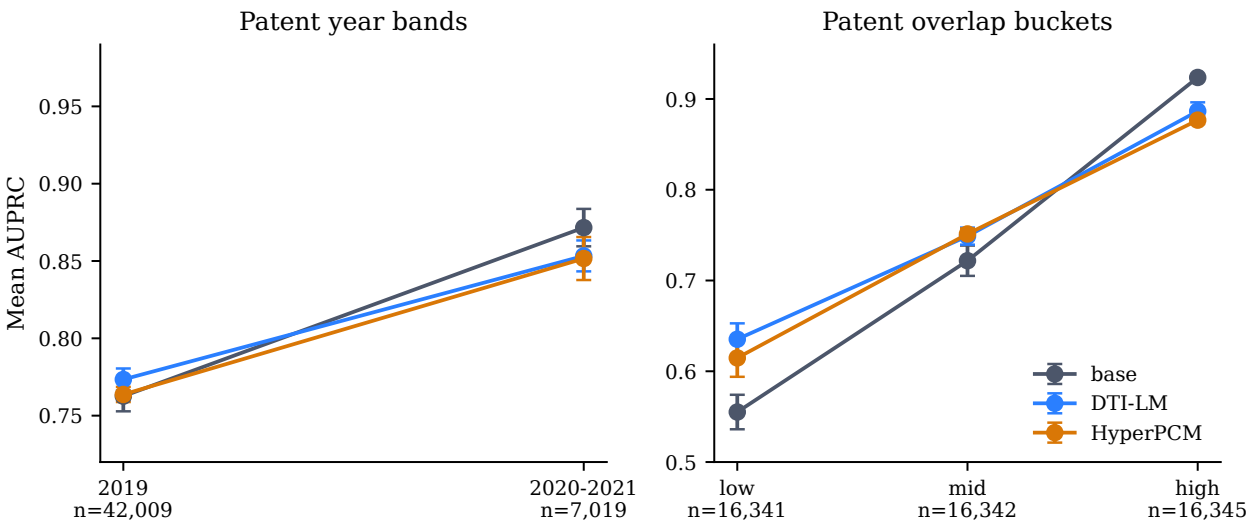

Figure 2: Patent temporal/provenance diagnosis in terms of mean AUPRC under the external replacement panel. Left: pooled year-band slices, with 2021 merged into the recent stratum. Right: test-set overlap buckets computed from the saved overlap score. Error bars denote seed std over the matched 3-seed runs. The aggregate `DTI-LM` advantage is concentrated in the 2019 and lower-overlap strata, while `base` dominates the recent and highest-overlap slices.

The `DAVIS` result is the closest supporting-panel case: `DTI-LM` is effectively tied with `base` on mean AUPRC and slightly higher on mean AUROC, but it still does not create a clean winner change of the kind seen on the promoted patent benchmark. `BindingDB_Ki` is more decisive, with `base` ahead of both external baselines on both metrics. These auxiliary results are therefore consistent with the main claim of the paper, but they do not carry the same argumentative weight as the patent benchmark because they do not add a cleaner ranking-reversal axis than temporal/provenance shift.

5 Discussion

The main message of this study is that, within the studied three-model panel, DTI evaluation is benchmark-sensitive in a way that current reporting practice often obscures. A synthetic split can still be useful, but it should not be treated as a universal proxy for real OOD generalization. Even within this compact panel, the sign of a method's apparent gain can change when the benchmark changes.

An important design choice in this paper is the use of a compact external model panel. A larger model zoo could be added later, but it is not necessary to establish the narrower claim studied here. The winner change already appears with three representative systems, and the shift-axis diagnosis helps explain where that winner change comes from. At the same time, the patent margin is small in absolute AUPRC, so the present paper should not be read as claiming a large practical deployment gap between `DTI-LM` and `base`. The safer interpretation is that a benchmark-qualified ranking reversal exists within this panel.

The paper also has clear limitations. Although the support benchmarks `BindingDB_Ki` and `DAVIS` remain useful counterexamples, they are still supporting rather than co-equal headline axes. The expanded model family is stronger than the original in-repo probe panel, but it is still partial rather than exhaustive: `DTI-LM` and `HyperPCM` do not cover the full space of recent DTI architectures. We now add two targeted robustness probes, and both make the manuscript more disciplined. The non-patent temporal benchmark shows that the `DTI-LM` edge is not confined to patent provenance alone, while the scaffold-drug probe shows that the original synthetic `unseen_drug` split was chemically permissive enough that the earlier `HyperPCM` edge does not survive a stricter holdout. Even with these additions, the patent benchmark is still temporal/provenance OOD rather than a pure time-only shift, so the current paper should not claim

that time alone explains the observed ranking change. A further caveat is that ChemBERTa and ESM2 were pretrained on broad corpora that may already include molecular and protein entities appearing in our temporal test sets. We therefore cannot rule out that part of the `DTI-LM` edge on temporal benchmarks reflects broader pretraining coverage of chemical or protein space rather than temporal robustness per se. We leave a mechanistic explanation for why pooled-LM features help more on temporal panels to future work, rather than claiming one from the current evidence alone. We also do not evaluate downstream biological decision quality. This paper should therefore be read as a well-supported benchmark case study rather than an exhaustive census of all real-OOD DTI settings. These limitations should be stated plainly, but they do not weaken the more disciplined central finding that synthetic and real-OOD rankings can disagree in the tested panel.

6 Conclusion

We presented a benchmark-first study of distribution shift in DTI prediction. A 3-seed synthetic reference panel on `BindingDB_Kd` showed that cold-start regimes are already heterogeneous even under an external literature-based comparison panel: `HyperPCM` is strongest on the original synthetic `unseen_drug`, `blind_start` has no reliable winner, and `base` remains strongest on synthetic `unseen_target` by mean AUPRC. Review-driven robustness checks then made the split semantics sharper: a scaffold-drug probe removed the original `HyperPCM` edge, and a non-patent temporal probe preserved the `DTI-LM` advantage beyond the patent benchmark itself. A promoted 3-seed temporal/provenance benchmark on `BindingDB_patent` still favored a different model family, with `DTI-LM` becoming the strongest model by mean AUPRC, albeit by a small absolute margin. Pooled patent year-band and overlap-bucket analyses further showed that real OOD is structured rather than monolithic. Overall, these results indicate that, within the studied three-model panel and tested benchmarks, synthetic cold splits and real distribution shifts should not be treated as interchangeable evaluation settings. This manuscript should be read as a benchmark/resource case study rather than an exhaustive field-wide census. Future DTI papers should therefore report both synthetic and real-OOD evidence when making generalization claims.

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A Appendix

A.1 Patent benchmark specification

Because the main paper depends on `BindingDB_patent / patent_temporal`, we make its construction explicit here. Table A1 enumerates the year boundaries, split counts, label construction, overlap policy, and seed semantics for both the main patent split and the earlier-cutoff robustness variant.

A.2 Non-patent temporal probe specification

Because the main revision also relies on `BindingDB_nonpatent_Kd / nonpatent_temporal` to separate temporal drift from patent provenance, we make that construction explicit as well. Table A2 lists the source file, year cutoffs, label transform, prevalence, duplicate policy, and seed semantics for the released non-patent temporal split.

Table A1: Explicit specification of the promoted patent benchmarks. The temporal partition is fixed by year; the random seed changes model stochasticity, not the partition itself.

Aspect	`patent_temporal`	`patent_temporal_v2017`
Source files	`train_val.csv + test.csv`	`train_val.csv + test.csv`
Train years	2013–2017	2013–2016
Validation year	2018	2017
Test years	2019–2021	2018–2021
Rows (train / valid / test)	142,275 / 41,155 / 49,028	90,771 / 51,504 / 90,183
Pos. rate (train / valid / test)	0.566 / 0.534 / 0.575	0.568 / 0.562 / 0.556
Label transform	raw patent value $Y \mapsto \mathrm{p}K_d = 9 - Y/\ln(10)$; positive label if $\mathrm{p}K_d \geq 7.0$	
Overlap policy	no cold-entity filtering; entity overlap is allowed, exported, and diagnosed via `overlap_score` buckets	
Duplicate policy	pair rows are kept after year partition; entity export tables are deduplicated by canonical ID	
Seed semantics	partition is fixed across seeds; seed affects training / initialization only	
OOD interpretation	temporal/provenance OOD rather than pure time-only shift	

Table A2: Explicit specification of the non-patent temporal probe. This benchmark is built from the official BindingDB TSV after filtering to non-patent, positive-K_d, single-chain rows with a valid publication year. The partition is fixed by publication year; the random seed changes model stochasticity, not the split.

Aspect	`nonpatent_temporal`
Source file	official `BindingDB_All_202603_tsv.zip`, filtered to empty `Patent Number`
Date field	publication year from `Date of publication`
Train years	1977–2018
Validation year	2019
Test years	2020–2023
Rows (train / valid / test)	35,755 / 3,006 / 7,174
Pos. rate (train / valid / test)	0.341 / 0.445 / 0.426
Label transform	raw K_d in nM with $\mathrm{p}K_d = -\log_{10}(Y/10^9)$; positive label if $\mathrm{p}K_d \geq 7.0$
Overlap policy	no cold-entity filtering; `drug_overlap`, `target_overlap`, and `overlap_score` are exported for diagnosis
Duplicate policy	exact duplicate rows are removed before partitioning; no exact duplicates remain in the released split files
Seed semantics	partition is fixed across seeds; seed affects training / initialization only
OOD interpretation	non-patent temporal shift; cleaner than patent provenance, but still not a pure time-only intervention

A.3 Reference-model specification

Because every ranking claim is anchored against `base`, we make that system explicit rather than treating it as an unnamed baseline. Table A3 specifies the exact input features, network structure, parameter count, optimization recipe, and implementation path used in every experiment.

A.4 Additional real-OOD support benchmarks

The main paper focuses on the strongest evidence chain: the synthetic `BindingDB_Kd` reference panel, the 3-seed `BindingDB_patent` ranking reversal, and the patent year/overlap diagnosis. `BindingDB_Ki` and `DAVIS` remain supporting real-OOD benchmarks: both now use matched 3-seed evaluation, but they reinforce rather than replace the patent temporal story.

On `BindingDB_Ki / unseen_target`, Table A4 shows that `base` remains ahead of both `DTI-LM` and `HyperPCM`. On `DAVIS / unseen_target`, `base` and `DTI-LM` are nearly tied on mean AUPRC, but `base` keeps the slightly stronger mean AUPRC and therefore remains the nominal winner for the paper's primary ranking metric. Together, these support panels reinforce the main paper's message that strong recent baselines do not transfer uniformly across OOD definitions.

Table A3: Specification of the `base` reference model used throughout the benchmark. This table is included to make the anchor system reproducible rather than treating it as an informal baseline label.

Component	Specification
Drug input	RDKit Morgan fingerprint, radius 2, 2048 bits, generated from canonical SMILES in `src/raicd/featurizers.py`.
Target input	L2-normalized hashed amino-acid 1/2/3-mer count vector, 2048 dimensions, generated in `src/raicd/featurizers.py`.
Drug encoder	Two-layer MLP: $\mathrm{Linear}(2048, 512)$, GELU, dropout 0.1, $\mathrm{Linear}(512, 256)$, followed by L2 normalization.
Target encoder	Same architecture as the drug encoder, with independent parameters.
Pair head	Concatenate $[d,\ t,\ d \odot t,\ \lvert d - t \rvert] \in \mathbb{R}^{1024}$, then $\mathrm{Linear}(1024, 512)$, GELU, dropout 0.1, $\mathrm{Linear}(512, 1)$.
Trainable parameters	2,886,145 parameters in total.
Optimization	BCE-with-logits objective, Adam with learning rate 10^{-3} and weight decay 10^{-4}, 8 epochs, batch size 256.
Labeling	Binary label derived from a `pKd` threshold of 7.0 (or patent-specific log-nM conversion for `BindingDB_patent`; see Table A1).
Code path	Model implementation in `src/raicd/model.py`; training entry point in `scripts/train_raicd.py`.

Table A4: Secondary real-OOD support benchmarks under the external replacement panel. These panels use matched 3-seed evaluation, but remain supporting evidence because they reinforce rather than replace the patent temporal benchmark.

Benchmark	base (AUPRC/AUROC)	DTI-LM (AUPRC/AUROC)	HyperPCM (AUPRC/AUROC)	Winner
`BindingDB_Ki / unseen_target`	**0.6934 ± 0.0257 / 0.7192 ± 0.0068**	0.6648 ± 0.0177 / 0.6936 ± 0.0044	0.6515 ± 0.0301 / 0.6757 ± 0.0211	base
`DAVIS / unseen_target`	**0.4097 ± 0.0275 / 0.8613 ± 0.0153**	0.4090 ± 0.0570 / 0.8718 ± 0.0152	0.3840 ± 0.0269 / 0.8610 ± 0.0245	base

A.5 Review-driven robustness probes

The external review highlighted two concrete risks: that the promoted patent result might be driven mainly by provenance rather than temporal shift, and that the synthetic `unseen_drug` split might still be chemically too permissive. Table A5 records the targeted fixes. The first probe constructs a non-patent temporal benchmark directly from the official BindingDB release and shows that `DTI-LM` still stays ahead of `base` over matched 3-seed evaluation. The second probe replaces entity holdout with a Murcko-scaffold holdout and shows that the original synthetic `HyperPCM` edge does not survive this stricter split.

A.6 Paired uncertainty support

Table A6 summarizes the paired bootstrap intervals for the main external-panel comparisons against `base`. The most important headline claims keep the same sign under paired resampling: `HyperPCM` is reliably stronger than `base` on synthetic `unseen_drug`, and `DTI-LM` is reliably stronger than `base` on the promoted patent temporal benchmark. The remaining main-panel comparisons either stay negative or cross zero, which is exactly why the paper frames them as split-specific rather than universal advantages.

A.7 Recent baseline screening and promotion

To avoid cherry-picking concerns, Table A7 summarizes the seed0 screening pass used to decide which recent baselines to promote. The promotion criterion was simple: prioritize the strongest recent model on the primary real-OOD benchmark `patent_temporal`, then keep any additional recent baseline that is competitive enough to test whether the main conclusion changes.

Table A5: Targeted robustness probes added after the external review. The non-patent temporal probe tests whether the promoted patent ranking persists when patent provenance is removed. The scaffold probe tests whether the synthetic `unseen_drug` ordering survives a stricter chemistry-aware holdout. Values are mean $\pm$ std over three seeds.

Benchmark	Systems	Mean AUPRC / AUROC	Paired AUPRC CI	Read
`BindingDB_nonpatent_Kd / nonpatent_temporal`	`base` vs `DTI-LM`	`base`: 0.6369 $\pm$ 0.0173 / 0.7333 $\pm$ 0.0081; `DTI-LM`: **0.6531 $\pm$ 0.0066 / 0.7644 $\pm$ 0.0041**	+0.0163 [+0.0064, +0.0265]	temporal ranking reversal survives without patent provenance
`BindingDB_Kd / scaffold_drug`	`base` vs `HyperPCM`	`base`: **0.6069 $\pm$ 0.0161 / 0.8351 $\pm$ 0.0037**; `HyperPCM`: 0.5983 $\pm$ 0.0368 / 0.8155 $\pm$ 0.0204	−0.0086 [−0.0203, +0.0025]	stricter scaffold holdout does not preserve the original `HyperPCM` edge

Table A6: Paired bootstrap support for the main external-panel comparisons against `base`, plus the review-driven robustness probes. Positive deltas favor the second model in the comparison; negative deltas favor `base`.

Comparison	Mean AUPRC delta	95% CI
`unseen_drug`: `DTI-LM - base`	−0.0505	[−0.0614, −0.0399]
`unseen_drug`: `HyperPCM - base`	+0.0185	[+0.0082, +0.0279]
`blind_start`: `DTI-LM - base`	−0.0100	[−0.0304, +0.0132]
`blind_start`: `HyperPCM - base`	+0.0092	[−0.0134, +0.0317]
`scaffold_drug`: `HyperPCM - base`	−0.0086	[−0.0203, +0.0025]
`unseen_target`: `DTI-LM - base`	−0.0088	[−0.0206, +0.0021]
`unseen_target`: `HyperPCM - base`	−0.0269	[−0.0390, −0.0151]
`patent`: `DTI-LM - base`	+0.0054	[+0.0028, +0.0079]
`patent`: `HyperPCM - base`	−0.0019	[−0.0045, +0.0009]
`nonpatent_temporal`: `DTI-LM - base`	+0.0163	[+0.0064, +0.0265]

A.8 Reusable benchmark resources

The benchmark release is intended to be reused independently of the in-repo models. Table A8 lists the concrete artifacts exported for each split and how they interact with the standalone evaluator. In practice, an external user only needs the canonical split bundle plus a prediction CSV keyed by `example_id`; the evaluator then reproduces the same global metrics and overlap-bucket metrics reported in this paper.

A.9 Method-centric ablations

Earlier project stages explored routers, conservative target-memory variants, and selective fallback rules. These analyses were useful for method diagnosis, but they do not define the final paper narrative. The benchmark paper is intentionally centered on ranking instability across shift definitions rather than on proposing a new best-performing method.

Table A7: Transparent seed0 screening used to decide which recent baselines were promoted to matched three-seed evaluation. The primary promotion criterion was performance on `BindingDB_patent / patent_temporal`, the main real-OOD benchmark.

Model	`patent_temporal`	`blind_start`	`unseen_target`	Promotion	Rationale
`DTI-LM`	0.7897 / 0.7435	0.3873 / 0.7130	0.5308 / 0.8011	main 3-seed promotion	strongest seed0 recent baseline on the primary real-OOD benchmark; therefore promoted in the main text
`HyperPCM`	0.7686 / 0.7247	0.3896 / 0.7447	0.5294 / 0.7990	secondary 3-seed promotion	competitive on synthetic seed0 axes, but weaker on the main patent benchmark; promoted only to test whether the main conclusion changes

Table A8: Reusable benchmark artifacts released with each split. The release is designed so that external predictions can be scored without re-running the in-repo training code.

Artifact	Example files	Purpose
Split manifest	`manifest.json`	versioned metadata, split counts, prevalence, seed, and canonical paths
Partition tables	`train.csv`, `valid.csv`, `test.csv`	canonical labeled partitions with `example_id` and exported overlap scores
Pair tables	`train_pairs.csv`, `test_pairs.csv`	lightweight pair-only views for external model pipelines
Entity tables	`drugs.csv`, `targets.csv`	unique drug / target vocabularies aligned to the exported split IDs
Full joined table	`all.csv`, `all_pairs.csv`	complete resource snapshot for auditing or custom slicing
Standalone evaluator	`scripts/evaluate_prediction_csv.py`	scores a prediction CSV against the canonical reference split and returns global + overlap-bucket metrics